

COE Applied AI Workshop		
Monday July 20		
Time	Activity	Lead
1:00 pm - 2:00 pm	From Computational Graphs to Auto-Differentiation	Edgar Lobaton
2:00 pm - 2:30 pm	Multi-Layer Perceptron: Foundations and Capabilities	Edgar Lobaton
2:30 pm - 3:00 pm	Hands-On: Exploring PyTorch and Autodifferentiation	Edgar Lobaton
3:00 pm - 3:15 pm	Coffee Break	
3:15 pm - 3:45 pm	Building from Scratch: Convolutional Neural Network (CNN)	Edgar Lobaton
3:45 pm - 4:15 pm	Hands-On: Exploring MLPs and CNNs	Edgar Lobaton
4:15 pm - 5:00 pm	Demo: Setting up a Local LLM	Tarek Aziz
Tuesday July 21		
Time	Activity	Lead
1:00 pm - 1:30 pm	Building from Scratch: Recurrent Neural Network (RNN)	Edgar Lobaton
1:30 pm - 2:00 pm	Building from Scratch: AutoEncoder (AE)	Edgar Lobaton
2:00 pm - 2:30 pm	Transfer Learning and Pre-Trained Models	Edgar Lobaton
2:30 pm - 3:00 pm	Hands-On: Exploring RNNs and AEs	Edgar Lobaton
3:00 pm - 3:15 pm	Coffee Break	
3:15 pm - 3:45 pm	Exploring DNN Architectures	Edgar Lobaton
3:45 pm - 4:15 pm	Hands-On: Transfer Learning	Edgar Lobaton
4:15 pm - 5:00 pm	Demo: Setting up a RAG	Tarek Aziz
Wednesday July 22		

Time	Activity	Lead
1:00 pm - 2:00 pm	Optimization and Regularization	Edgar Lobaton
2:00 pm - 2:30 pm	Generative Adversarial Network (GAN)	Edgar Lobaton
2:30 pm - 3:00 pm	Transformers	Edgar Lobaton
3:00 pm - 3:15 pm	Coffee Break	
3:15 pm - 3:45 pm	Diffusion Models	Edgar Lobaton
3:45 pm - 4:15 pm	From Generative Models to AI Assistants	Edgar Lobaton
4:15 pm - 5:00 pm	Demo or Hands-On	Edgar Lobaton